



GE Healthcare

Technical Publications

Direction 5158952-100
Revision 5

Functool 6 Installation/Service Manual

Copyright © 2007 by General Electric Company Inc.
All Rights Reserved

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Contact your local service coordinator for more information on this process.

Rev. 06/09/2006

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

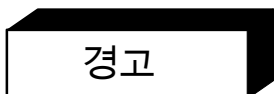
- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・このサービスマニュアルは英語版しかありません。
- ・GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- ・このサービスマニュアルを熟読し、理解せずに装置のサービスを行わないでください。
- ・この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意:

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。



.....
.....,
.....
.....
.....,
.....

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
1..	September 20, 2005	Initial Version
2...	November 21, 2005	Add FRU Part Number to Section 7
3...	June 09, 2006	Add Installation finish tasks as Section 6 and Update FRU Part Number in Section 8
4...	January 15, 2007....	Updated FRU part number in section 8
4...	June 14, 2007.....	Added Korean warning above, changed cd/ to cd / in section 2-2-2, updated FRU part number in section 8
5...	November 14, 2007	Upgraded to Functool 6, removed floppy disk content, updated GIB website address

PAGE	REV	PAGE	REV	PAGE	REV	PAGE	REV
Title Page.....	5	5	4	9	3		
2	2	6	2	10	5		
3to 4	1	7	1				
		8	5				

* This revision/letter corresponds to the indicated document's revision control system.

TABLE OF CONTENTS

1- INTRODUCTION	7
1-1 Functool 6 Advanced Application “Clinical” Summary	7
2- PRELIMINARY REQUIREMENTS	7
2-1 License Key Number	7
2-2 Save Configuration	8
2-2-1 Reading Config Media → PC.....	8
2-2-2 Reading Config Media → AW	8
3- INSTALLATION PROCEDURE	8
4- AW AND FUNCTOOL 6 ERRORS.....	8
5- ADVANCED SOFTWARE APPLICATION OPERATION AND DEMO EXAM INFORMATION	8
6- INSTALLATION FINISH TASKS	9
7- SPECIAL CONSIDERATIONS.....	9
8- FRU LIST.....	10

1- INTRODUCTION

The following manual gives a brief overview of the installation process for the Functool 6 advanced application software option. The Functool 6 application is designed to run on the AW4.1 and forward AW platforms only. This guide makes references to all the appropriate standard Advantage Workstation service manuals for more detailed information about installing an advanced application software option on the AW. The current revision of the appropriate AW Service Manual should be used when installing this software. Unique installation processes for Functool 6 and NOT found in the AW service manual can be found in this document under the section entitled SPECIAL CONSIDERATIONS.

1-1 Functool 6 Advanced Application “Clinical” Summary

Functool 6 is an application that is used to analyze Magnetic Resonance Imaging (MRI) data sets that contain multiple images for each scan location, representing changes in intensity over time (“time series” or “dynamic” data sets), b-value (diffusion imaging), gradient orientation (diffusion tensor imaging) or frequency (spectroscopy).

Functool 6 provides the user with two main tools for analysis:

- ❑ Graphs – these show the changes in pixel value over time, b-value or frequency at a given point on the images, either at the cursor position or within a region of interest (ROI) defined by the user.
- ❑ Functional maps – the pixel values from the data set are used to compute a characteristic parameter for every pixel location by means of a function. This can be a basic algorithm or a function derived from such an algorithm. A functional map is then constructed by displaying the value of this parameter for every pixel location.

Functool 6 is supplied in the form of protocols. Each consists of one or more protocol panels that contain the necessary information and the controls required to set up an analysis.

2- PRELIMINARY REQUIREMENTS

2-1 License Key Number

The Functool 6 option requires a license key # based upon the workstation’s Host ID. This key number may be provided by:

- ❑ Paperwork shipped with the option that will include one or more 8 to 10 digit license key number(s) – created based on the workstation’s/application’s FDO and the workstation’s Host ID/License ID.
- ❑ The software package may include a portable media (i.e. floppy disk) with the license key information on it.
- ❑ The license key information will also be available online at the eLicense website link by searching under the order number or the workstation’s Host ID/License ID: <http://egems.gemedicalsystems.com/elicense/index.jsp>.

All of these elements may exist (hard copy license #(s), media, and eLicense data), and at least one of them is necessary to load the software.

2-2 Save Configuration

Create an AW 'CONFIG' media before and after the application software installation for use as needed throughout this and other AW service activities.

2-2-1 Reading Config Media → PC

Configuration information can be read on a PC by viewing the config.txt file on the media using text readers like WordPad, Internet Explorer or MS Word (MS NotePad may not work).

2-2-2 Reading Config Media → AW

To read a Save Configuration using a CDROM disk, insert the CD and type the following in a c shell:

- **su**
- Password '**operator**'
- **mount /mnt/cdrom**
- **cd /mnt/cdrom**
- **more ("backup directory name")**
- **cd /**
- **umount /mnt/cdrom**

3- INSTALLATION PROCEDURE

The 'Standard' AW Software Options Installation – Easy Install process applies. For ongoing process revision viability, the details will not be printed here, but they can be found in the AW Service Manual.

The current revisions for this manual can be viewed online at: <http://aw-ib.euro.med.ge.com>.

1. Open up the appropriate Advantage Workstation (AW) Service Manual.
2. Proceed to Job Card entitled – Software Options Installation.
3. Follow the procedure using the "Easy Install" method.
4. When complete, perform a save configuration.

4- AW AND FUNCTOOL 6 ERRORS

Funtool 6 and AW System errors can be examined and investigated in either the usual AW error logs – such as /var/adm/messages, and/or ~sdc/logfiles/EasyInstallLog. Errors specific to Funtool 6 can be found in /export/home/sdc/logfiles/functool.log.

5- ADVANCED SOFTWARE APPLICATION OPERATION AND DEMO EXAM INFORMATION

To test the functionality of the software, first you must load the DEMO exam CD shipped with the product. The 'Standard' AW Software Demo Exam Installation process applies. For ongoing process revision viability, the details will not be printed here, but they can be found in the AW Service Manual.

The current revisions for this manual can be viewed online at: <http://aw-ib.euro.med.ge.com>.

1. Open up the appropriate Advantage Workstation (AW) Service Manual.
2. Proceed to Job Card entitled – Advantage Workstation Demo Exam Reinstallation.
3. Follow the procedure to load the Demo Exams from the CD provided.

To test this application, after the application is installed, logout and log back into the AW (as suggested by the messages after the installation). Then select any data set and click on Functool in the “Applications” menu. The Functool window will open (to the left side of the screen). This ensures that Functool is installed correctly. Click on the Quit button to exit Functool.

6- INSTALLATION FINISH TASKS

GIB Update - The Functool6 application must be entered into GIB (Global Installed Base) database after the initial successful installation in order to fulfill product-tracking requirements.

FAILURE TO ENTER THE SYSTEM AND ALL SOFTWARE APPLICATIONS INTO GIB will result in GE Healthcare being in violation of FDA tracking compliance, and this system will not be in the affectivity list for future upgrades.

The OLC AW Support line can direct you as how to get GIB updated correctly if you don't know. Here is the global GIB link:

GIB: <http://gib.gehealthcare.com>

7- SPECIAL CONSIDERATIONS

In addition to Functool, Functool CSI, Functool DTI and Functool FiberTrak will show up at the package installation time as a separate license key entry.

If Functool CSI and/or Functool DTI and/or Functool FiberTrak was purchased, and have license keys, they must be selected and the license key numbers must be entered at the installation prompt for it – otherwise leave it unselected.

There can be up to four license keys with this option

- o Functool **MUST** exist
- o Functool CSI may exist
- o Functool DTI may exist
- o Functool FiberTrak may exist

8- FRU LIST

The replacement parts for this option are as follows:

Part#5160880-6 – Functool6 Software and Documentation CD Rom